Meibomianitis and Associated Skin Disorders

Lorne Yudcovitch, OD, MS, FAAO

1

SOAP Format

- S = Subjective
  - AKA symptoms
  - Everything the patient says
- O = Objective
  - AKA signs
  - All the doctors observes and all the test results
- A = Assessment
  - AKA diagnosis (or diagnoses)
  - What is wrong with the patient
- P = plan
  - For each diagnosis, what is going to be done to help the patient

2

Meibomian Gland Dysfunction

- Not considered an infectious entity
- Altered meibomian gland metabolism
- → secretions become thickened and block the gland opening
- → stagnant material becomes growth medium for bacteria
- → bacteria produce lipase and the released fatty acids trigger inflammation

3

Meibomian Gland Dysfunction

Often associated with (acne) rosacea

- **Etiology**
  - **Chronic inflammatory disorder involving skin and eye**
  - **Sebaceous glands**
- **Epidemiology**
  - **"Celtic", northern ancestry (genetic component)**
  - **Adults 30-60 years of age**
  - **> 30% have ocular problems**

4

Rosacea

- **Redness (erythema) in the "facial flush area"**

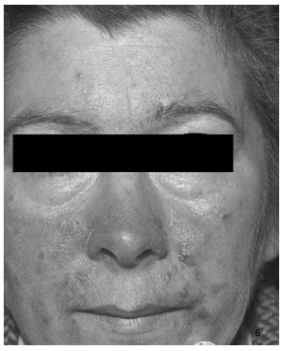


5

Rosacea

- **Papules and pustules in the "flush areas"**



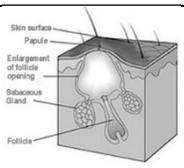


6

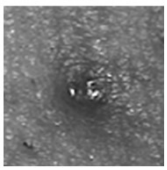
Rosacea

Papule

- Red and inflamed lesion on the skin's surface
- Contains no pus or white head
- High break in the follicle wall
- Cellular debris and bacteria spill into the dermis.
- Often progresses to a pustule



Papule

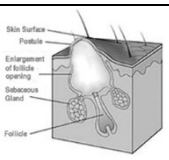


7

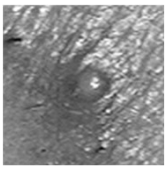
Rosacea

Pustule

- A typical 'pimple'
- Red and inflamed with a white head
- Sometimes brownish core can be seen as a cap on the pus head
- Pustules follow papules after white blood cells invade
- White cap on a pustule is pus, sebaceous matter and cellular debris



Pustule



8

Rosacea

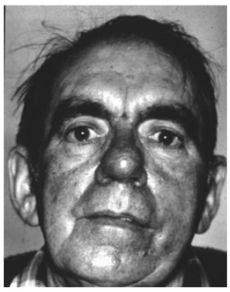
- Telangiectasias - visible (dilated) small blood vessels on nose or cheeks



9

Rosacea

- Rhinophyma - red, bullous nose (sebaceous gland hypertrophy)



10

- W.C. Fields 1880 – 1946



11

Rosacea

Treatment of facial rosacea

- Doxycycline 100 mg bid x 4-6 weeks
- Topical metrodiazole gel 0.75% bid x 5-8 weeks



12

Rosacea

Treatment of Facial Rosacea



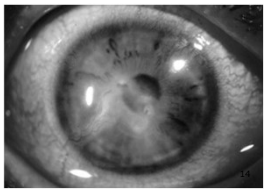
- Dietary restrictions that trigger reaction
  - well-known rosacea tripwires:
    - hot spices, alcohol, heated beverages
  - lesser-known rosacea tripwires:
    - chocolate, tomatoes, citrus fruits
  - No significant effect:
    - dairy products (milk, cheese)
- Environmental restrictions that may trigger reaction
  - Sun exposure, strong winds, strenuous exercise, change in humidity/weather
  - Emotional factors (stress, fear, anxiety, embarrassment)
  - Heliobacter pylori (bacteria linked with stomach ulcers)
  - Vasodilators

13

Ocular Manifestations of Rosacea

- Meibomian gland dysfunction
- Blepharitis
- Hordeolum
- Chalazion
- Inflammation and infection of the lids
- Poor tear film
- Conjunctival injection
- PEE (SPK)
- Corneal vascularization
- Corneal thinning





14

Rosacea

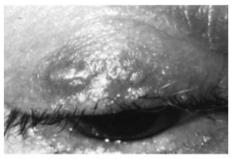
- Papules and pustules
- Telangiectasias
- Staph blepharitis
- Madarosis
- Meibomian gland dysfunction
- Conjunctivitis

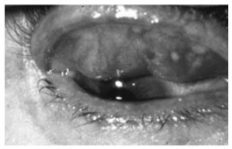


15

Ocular Rosacea

- Infection and inflammation of the lids

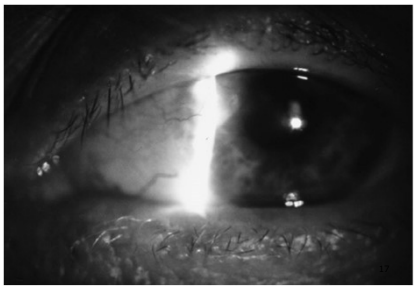




16

Ocular Rosacea

- Keratitis
  - Corneal vascularization and thinning



17

Ocular Rosacea

Treatment

- Meibomian gland dysfunction
  - Warm compresses, lid scrubs, gland expression
- Blepharitis
  - Same as staphylococcal blepharitis
- Keratitis
  - Artificial tears/ointments
  - Steroids if severe inflammation



18

Ocular Rosacea

- Follow-up
  - Without corneal involvement - weeks
  - With corneal involvement - days
- Patient education
  - Lid hygiene
  - Dietary restrictions (triggers)
  - Environmental triggers



19

Meibomian Gland Dysfunction

- Normal meibomian glands



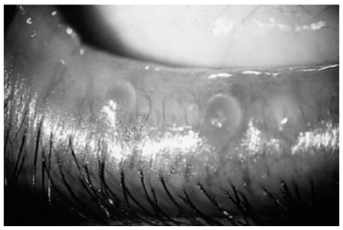
- Abnormal meibomian glands



20

Meibomian Gland Dysfunction

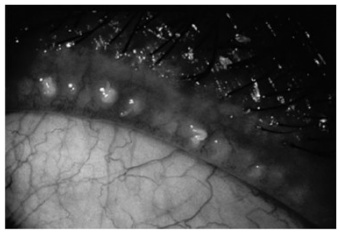
- Plugged meibomian glands
- Meibomian 'caps'



21

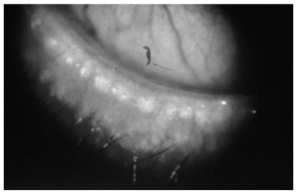
Meibomian Gland Dysfunction

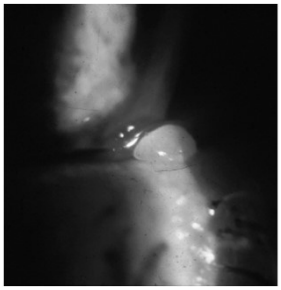
- Significant pouting of meibomian glands
- Inflammation of lid surrounding glands
- Serrated lid margin
- Tarsal conjunctival streaking
- Conjunctival injection



22

Meibomian Gland Dysfunction

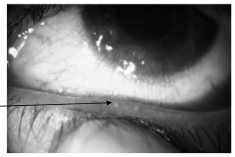


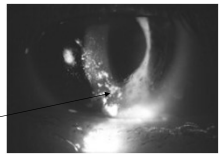


23

Meibomian Gland Dysfunction

- Diagnosis usually made clinically
- Forced expression of the meibomian glands
  - "Toothpaste" test - thickened secretions are forced out of some glands
  - Many glands will yield no visible secretion
  - Thickened secretions in the tears





24

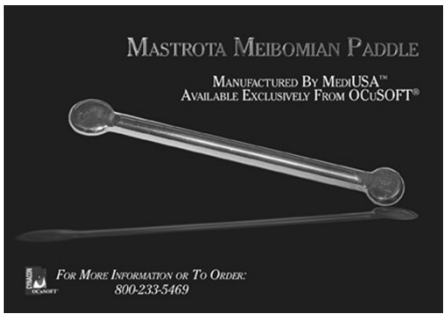
Meibomian Gland Dysfunction

Treatment

- **Warm compresses**
  - **Softens any material on the meibomian gland orifices**
  - **Softens thickened secretions in glands**
- **In-office and at-home meibomian gland expression (digital massage)**
  - **Pressure on the lids from the part of the tarsus away from lid margin towards lid margin (opening)**
  - **Open the ducts of the glands and express the thickened secretions**

25

Meibomian Gland Dysfunction



26

Meibomian Gland Dysfunction

Treatment

- **If disease severe or not responding to therapy**
  - **Consider doxycycline 100 mg bid for 6 to 10 weeks**
  - **Taper to lowest dose or, if possible, discontinue**
  - **Works by inhibiting lipase production**



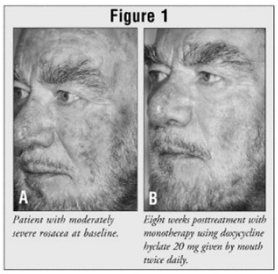
- **Manage dry eye**
- **Evaluate overlap with staphylococcal blepharitis**
  - **Treat both conditions**
- **Evaluate overlap with seborrheic blepharitis**
  - **Treat both conditions**
- **Evaluate if rosacea present**
  - **Consider dermatologic consult**



27

Meibomian Gland Dysfunction

Follow-up
- **1 week**
- **Then 3-4 weeks later**

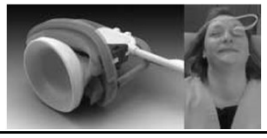


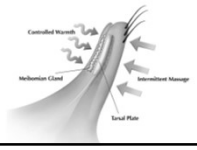
Patient education
- **Chronic condition**
- **No cure, management only**

28

LipiFlow Thermal Pulsation System (TearScience)

- **Meibomian expression in-office treatment device**
- **Invented by Donald Korb, O.D.**
- **Applies heat to tarsal side, pressure to skin side**
- **Can adjust heat/pressure; 12 minute sessions**
- **FDA-approved 7/11/11**





29

Intense Pulsed Light Therapy (IPL)



30

Intense Pulsed Light Therapy (IPL)

- Non-laser, high intensity light
- Hand piece contacts skin
- Intensity, duration and wavelengths vary based on skin type/treatment goals
- Thermal heating of meibum, superficial telangiectasia
- Potential benefit for ocular rosacea and dry eye related to meibomian gland dysfunction (MGD)
- Increases lipid layer and noninvasive TBUT



31

Impetigo

- Skin infection with *Staphylococcus aureus* or *Streptococcus pyogenes*
- Most common
  - Children under age 6
  - Sometimes very elderly
  - Exposed to poor hygienic conditions
- Contagious



32

Impetigo

- Rash with crusts ('honey crusts')
- Face, distal extremities



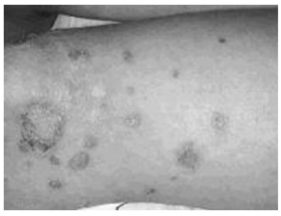




33

Impetigo

- Oral antibiotic
  - Culture and susceptibility testing
  - Cephalosporin
  - Macrolide
  - Dicloxacillin
- Wash involved areas
- Topical antibiotic after washing
  - Bacitracin
  - Mupirocin

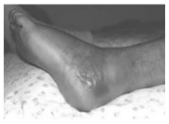


34

Erysipelas

- Skin infection involving the dermis
  - Superficial dermal lymphatics swell up
  - local redness, heat, swelling, and highly characteristic raised, indurated border
- Most common in young and elderly
- Legs and face most frequently affected sites
- Caused by *Streptococcus pyogenes*

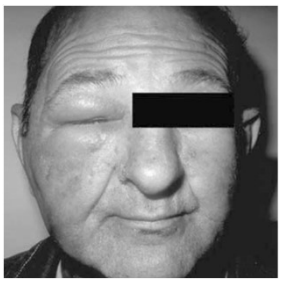




35

Erysipelas

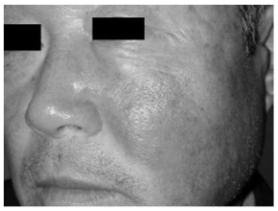
- Fever, chills
- Malaise
- Painful edematous patches on skin
- If on face, can continue into lids
- Sharply demarcated areas of swelling and redness



36

Erysipelas







- Treatment
  - PCP and/or dermatologic consult
  - Antibiotic: Penicillin G (injected) or penicillin V (PO)

THANK YOU!

38